

The HD Engine: Zero to Live

A guide for vibe coders, intuition-led builders, and digital architects.

Welcome. If you are reading this, you are about to spin up a complex astrological and Human Design calculation engine. Code can feel like an exclusionary wall of jargon, but at its core, it is just setting up a space, gathering the right ingredients, and speaking the right spell.

This guide assumes you know what a computer is, but not much else about programming. We are going to build your environment from the ground up so that the NeverSubtract HD engine runs flawlessly.

Phase 1: Opening the Canvas

Everything we do will happen in the **Terminal**. If you are on a Mac, open your applications, search for "Terminal," and open it. It will look like a blank, slightly intimidating text box. Take a breath. This is just a direct line of communication to your machine's deeper architecture.

THE GOLDEN RULE OF THE TERMINAL

In the terminal, you type a command and press **Enter**. You wait for the machine to finish its thought (which might look like walls of text scrolling by). When the prompt returns and stops moving, it is ready for your next command.

Phase 2: Finding Your Folder

The terminal is blind. It only knows about the exact folder it is standing in. Before we do anything, you must tell the terminal to walk into the folder where you saved the `chart.py` file.

Type `cd` (which stands for change directory), press the spacebar once, and then drag and drop the folder containing your code directly into the terminal window. It will fill in the path for you. Press Enter.

Phase 3: Homebrew (The Magic App Store)

Your Mac needs a way to download developer tools safely. We use a system called Homebrew. It is essentially an invisible app store for coders.

THE COPY-PASTE TRAP

When copying code from this document, **triple-click** the block so it highlights as one solid line. If you copy a command and it breaks into two lines when you paste it into the terminal, it will fail. Make sure it stays on one continuous line.

Copy this entire block, paste it, and press Enter. It may ask for your computer's password. When you type your password, *no characters will show up*. That is normal. Type it and hit Enter.

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Phase 4: Python (The Brain)

Macs come with a version of Python, but it is locked down. We need an unchained version.

```
brew install python
```

Phase 5: The Bubble (Virtual Environments)

If you install code packages directly into your main system, things break. We create a **Virtual Environment**—a safe bubble for our engine.

Create the bubble:

```
python3 -m venv hd_engine_env
```

Step inside the bubble. **You must do this every time you open a new terminal window to work on this project.**

```
source hd_engine_env/bin/activate
```

(You will know it worked because `(hd_engine_env)` will appear at the start of your terminal line.)

Phase 6: Gathering the Ingredients

Now we download the specific knowledge our script relies on: planetary ephemerides, time zone logic, and web server tools. *(Remember: triple-click to copy as one line!)*

```
pip install pyswisseph fastapi uvicorn pydantic geopy timezonefinder  
pytz
```

Phase 7: Awakening the Engine

You have the environment, the ingredients, and you are standing in the right folder. It is time to run it.

Mode A: The Local Oracle (Command Line)

To test a chart quickly and print it beautifully in your terminal:

```
python chart.py
```

Mode B: The Live API (For Web & Automations)

To connect this engine to a web interface (like Projector.Solutions) or n8n:

```
uvicorn chart:app --reload
```

Your engine is now live. Open a web browser and go to **http://127.0.0.1:8000/docs** to see the interface.